

Brian J. Sousa

Doctoral Student in Environmental Health

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Education

Boston University, PhD in Environmental Health – Boston, MA 2025 – present

Boston University, MS in Population Health Research, Climate and Health – Boston, MA 2023 – 2025

Northeastern University, BS in Applied Physics – Boston, MA 2015 – 2020

Experience

Research Analyst, Boston University School of Public Health – Boston, MA Aug 2023 – Aug 2025

- Updated and maintained the Co-Benefits of the Built Environment (CoBE) tool and assisted with development of future versions and modules added to the CoBE toolbox.
- Designed novel experiments run through the CoBE tool and assisted in the writing of papers intended for submission for the peer-reviewed literature.
- Reported and presented findings to outside stakeholders, including funders, policymakers, building owners, investors, and other stakeholders.
- Collected, cataloged, and QA/QC energy infrastructure and population health data.
- Used GIS analytical techniques to assess siting patterns of energy infrastructure.
- Developed exposure indices based on energy infrastructure data and data on emissions, use intensity, or other variables that serve as a proxy for intensity of use.
- Reported results and participated in workshops for the research community at BUSPH and the Institute for Global Sustainability at Boston University.

Research Assistant, Harvard T.H. Chan School of Public Health – Boston, MA May 2022 – July 2023

- Managed the CoBE Tool, a web-based tool to aid in understanding the climate and health impacts of the energy consumption of buildings.
- Assisted in updating the CoBE Tool and developing new tools and features.
- Conducted modeling and analysis for ongoing co-benefits research projects.
- Led independent research projects and presented findings at conferences.
- Developed case studies using the CoBE Tool and co-authored papers reporting findings.

Risk and Transaction Analyst, REsurety, Inc. – Boston, MA Jan 2021 – Apr 2022

- Settled hedging contracts for renewable energy projects by cleaning, and analyzing raw site data using R.
- Evaluated renewable energy contracts for utility scale projects using meteorological and electricity market data.
- Worked with other analysts to execute quality control checks on data critical to business processes.

Research Intern, National Renewable Energy Laboratory – Golden, CO Aug 2020 – Dec 2020

- Collected, parsed, and cleaned hourly emissions data for natural gas power plants in order to analyze nitrogen oxide to describe their emissions profiles empirically and statistically during categorized non-steady state conditions.
- Applied k-means and gaussian mixture model clustering algorithms using python to differentiate steady and non-steady state power plant emission rates.
- Designed and presented a research poster summarizing the project findings.

Research Assistant, Center for Climate, Health, and The Global Environment – Boston, MA Feb 2020 – Aug 2020

- Built a program in python to plot and test the reported emission rates by the EPA of SO₂, CO₂, and NO_x from multiple types of power plants in the Northeast.
- Ran simulations using GridPIQ and AVERT programs in order to test the difference in emissions of different charging scenarios for electric vehicles in regions throughout the US.
- Compared the emissions for multiple electric vehicles charging scenarios with the corresponding reduction in emissions from the removal of fossil fuel powered vehicles to estimate the total bounce back in emissions.

Solar Design Engineering Intern, Nexamp, Inc – Boston, MA June 2019 – Dec 2019

- Designed site plans for potential solar projects in AutoCAD, Helios, and Helioscope, while considering local and national standards for offsets and design limitations.
- Performed energy model simulations in PVSyst in order to analyze the capacity factor of projects and the losses due to shading and stringing.
- Worked with Design and Power Engineers to update later stage project designs based on updated drawings.

Awards

Graduate Student Summer Fellow Jan 2024

Presentations

- Sousa, B., Who Lives Near Crypto Mines? Characterizing Populations Near Cryptocurrency Mining Centers in the U.S. - Seminar presentation at the Gijs Van Seventer Environmental Health Seminar, Boston University School of Public Health, Department of Environmental Health. April 2026: Boston, MA.
- Sousa, B., Buonocore, J., Fabian, M. P., Ryan, E., & Cleveland, C., Quantifying the Health Co-Benefits of Air Pollutant Reductions From the Mass Save Program - Poster presented at the American Geophysical Union (AGU) Annual Meeting. December 2025: New Orleans, Louisiana.
- Sousa B., Buonocore J., Incorporating Health and Water Impacts into a Cost-Benefit Assessment of Climate Mitigation Technologies in the United States - Poster presented at the American Geophysical Union (AGU) Annual Meeting. December 2024: Washington, D.C.
- Sousa B., Willis M., van Loenen BD., Mooney F., Buonocore JJ., The [EI]3 Database: A Tool for Understanding the Health Impacts of U.S. Energy Infrastructure - Poster presented at the International Society of Exposure Science (ISES) Annual Meeting. October 2024: Montreal, Canada
- Sousa B., Buonocore J., Salimifard P., Lahvis M., Allen J., Understanding Seasonal Variations of U.S. State Building Energy Demand - Preliminary findings presented at the International Society for Environmental Epidemiology - North American Chapter (ISEE NAC) Conference. June 2023: Corvallis, Oregon.
- Sousa B., Buonocore J., Salimifard P., Parikh S., Lahvis M., Allen J., Understanding Seasonal Variations of U.S. State Energy Demand [GC42Q - Multi-sector Dynamics: Multi-sector Impacts of Energy Transitions II] - Poster presented at the American Geophysical Union (AGU) Fall Meeting. December 2022: Chicago, Illinois.

Publications

Direct air capture has substantial health and climate opportunity costs Jan 2026
Kashtan, Y., Pendleton, J., Sousa, B., Willis, M. D., Michanowicz, D. R., Shonkoff, S. B. C., Buonocore, J. J.
doi.org/10.1038/s44458-026-00068-0 (Communications Sustainability)

High populations near fossil fuel energy infrastructure across the supply chain and implications for an equitable energy transition Jan 2025
Buonocore, J. J., Mooney, F. A., Campbell, E. J., Sousa, B., Van Loenen, B., Fabian, M. P., Nori-Sarma, A., Willis, M. D.
doi.org/10.1088/1748-9326/ae0da6 (Environmental Research Letters)

Assessing the climate and health impacts of energy consumption in European Union countries Jan 2025
Pei, G., Buonocore, J. J., Salimifard, P., Sousa, B., Ferguson, L., Allen, J. G.
doi.org/10.1088/1748-9326/add752 (Environmental Research Letters)

Federal policy platforms and public health: Reinforcing the benefits of air pollution control devices at power plants in the United States

Jan 2025

Buonocore, J. J., Fisher, J., Prull, D., Willis, M. D., Arunachalam, S., Perera, F., Kinney, P., Sousa, B., Levy, J. I.
doi.org/10.2105/AJPH.2024.307896 (American Journal of Public Health)

A novel method for calculating the projected health and climate co-benefits of energy savings through 2050

Jan 2023

Salimifard, P., Rainbolt, M. V., Buonocore, J. J., Lahvis, M., Sousa, B., Allen, J. G.
doi.org/10.1016/j.buildenv.2023.110618 (Building and Environment)

Early Adoption of strategies to meet building performance standards increases benefits to both climate and health

Salimifard, P., Sousa, B., Buonocore, J., Lahvis, M., Pendleton, J., Rainbolt, M., Allen, J. G.
(Under Review)